CHANGE OF BASIS

$$
\begin{aligned}
|r\rangle &= \frac{1}{\sqrt{2}}|u\rangle + \frac{1}{\sqrt{2}}|d\rangle \\
|l\rangle &= \frac{1}{\sqrt{2}}|u\rangle - \frac{1}{\sqrt{2}}|d\rangle \\
\\
|i\rangle &= \frac{1}{\sqrt{2}}|u\rangle + \frac{i}{\sqrt{2}}|d\rangle \\
|o\rangle &= \frac{1}{\sqrt{2}}|u\rangle - \frac{i}{\sqrt{2}}|d\rangle
\end{aligned}
$$

SPIN COMPONENT IN THE $\hat{n}$ DIRECTION

Vector Notation

$$\sigma_n = \vec{\sigma} \cdot \hat{n}$$

Component Form

$$\sigma_n = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z$$

More Concretely

$$\sigma_n = n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Combined in a Single Matrix